MATH 3551 A — Abstract Algebra I
Practice Midterm 1

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For practice — Midterm 1 is Monday, October 12, 2026 • Time limit: 50 minutes • Total: 100 points

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NAME (please print): ______________________________________

Directions. This exam is **closed book**: no textbook, no notes, no calculators, no phones, no other devices, and no assistance of any kind. Write your answers on the exam itself; use the backs of pages if you need more room, and say clearly when you have done so. *Show your work and name the theorems you use* — "by Lagrange's Theorem, $|H|$ divides 24" earns nearly all of the credit, an unjustified correct number earns little. Point values are marked on each problem. Good luck.

Problem 1. (30 points) **True or false; prove or disprove.**

Circle **T** or **F**, and then give a one-line reason: a theorem, a short computation, or an explicit counterexample. Three points each: one for the circle, two for the reason.

(a) **T** **F** There is an element of order 18 in the symmetric group S_9.

(b) **T** **F** The map $\varphi\colon \mathbb{Q}^\times \to \mathbb{Q}^\times$ defined by $\varphi(x) = |x|$ is a group homomorphism.

(c) **T** **F** The center of the dihedral group D_{16} of order 16 is $\{1, r^4\}$.

(d) **T** **F** If $H \le K \le G$ and $|G : H|$ is finite, then $|G : H| = |G : K| \cdot |K : H|$.

(e) **T** **F** If every proper subgroup of a group G is cyclic, then G is cyclic.

(f) **T** **F** The set of elements of finite order in an abelian group G is a subgroup of G.

(g) **T** **F** The group $(\mathbb{Z}/15\mathbb{Z})^\times$ is cyclic.

(h) **T** **F** The number of 3-cycles in S_6 is 40.

(i) **T** **F** If $x, y \in G$ satisfy $|x| = |y| = 2$ and $xy = yx$, then $|xy| = 2$.

(j) **T** **F** The groups $\mathbb{Z}/4\mathbb{Z}$ and $(\mathbb{Z}/8\mathbb{Z})^\times$ are isomorphic.

Problem 2. (20 points) **Inside $\mathbb{Z}/72\mathbb{Z}$.**

Work in the additive group $\mathbb{Z}/72\mathbb{Z}$.

(a) (4 points) Find the order of $\overline{30}$.

(b) (5 points) List the elements of the subgroup $\langle\overline{30}\rangle$.

(c) (4 points) How many generators does $\mathbb{Z}/72\mathbb{Z}$ have?

(d) (7 points) Determine every k with $0 \leq k < 72$ for which $\langle\overline{k}\rangle = \langle\overline{18}\rangle$.

Problem 3. (15 points) **Two permutations.**

In S_9 let
$$\sigma = (1\,4\,7)(2\,5)(3\,6\,8\,9), \qquad \tau = (1\,2\,3\,4\,5).$$
(Permutations act on the left, so $\sigma\tau$ means: first apply τ, then apply σ.)

(a) (4 points) What is $|\sigma|$?

(b) (6 points) Find the cycle decomposition of $\sigma\tau$, and its order.

(c) (3 points) For which k with $0 \leq k < 12$ does σ^k fix the point 3?

(d) (2 points) Give the cycle decomposition of σ^{-1}.

Problem 4. (15 points) **Centralizer and normalizer in the quaternion group.**

Let $G = Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$, with the usual relations $i^2 = j^2 = k^2 = -1$, $ij = k$, $ji = -k$, and let $A = \langle i \rangle = \{1, -1, i, -i\}$.

(a) (6 points) Compute $C_G(A)$. Justify your answer.

(b) (6 points) Compute $N_G(A)$. Justify your answer.

(c) (3 points) List the left cosets of A in G and check Lagrange's Theorem for this pair.

Problem 5. (20 points) **The center.**

Let G be a group with center $Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}$.

(a) (5 points) Prove that $Z(G) \trianglelefteq G$.

(b) (10 points) Prove that if the quotient group $G/Z(G)$ is cyclic, then G is abelian.

(c) (5 points) Deduce that if G is a finite group, then $|G : Z(G)|$ is never a prime number.

End of practice exam. Total: $30 + 20 + 15 + 15 + 20 = 100$ points.